

Comparison of optical see-through and video see-through schemes in augmented reality display technology

Haibo Lu*, Deyou Xu, Qunfang Zhang, Fei Hou

Artillery Military Theory Innovation and Operation Experiment Center, Army Academy of Artillery and Air Defense, Nanjing 211132, China

* Corresponding author: lhbboo@sohu.com

ABSTRACT

Augmented reality (AR) technology presents real environments and digital information simultaneously to people in an unprecedented way, becoming a key direction for the development of current information technology and industry, as well as a research hotspot. Display technology is one of the keys to AR technology, and the choice of display scheme determines the quality of the fusion of reality and virtuality under different application conditions. There are two display schemes for augmented reality, the optical see-through (OST) and the video see-through (VST). In the paper, the imaging principles of two kinds of display schemes are introduced briefly. Comparisons were made of the two scheme in terms of real-world resolution, image quality, field of view (FOV), occlusion effect, brightness and brightness matching control, latency, registration information. The advantages and disadvantages of each scheme are discussed. The applicable fields and main directions for improvement of two schemes are analyzed. It is hoped that this paper can provide assistance in the selection and development of augmented reality display schemes.

Keywords: Augmented Reality, Optical See-Through, Video See-Through

1. INTRODUCTION

The earliest augmented reality (AR) system emerged in 1968 when Ivan Sutherland developed the world's first augmented reality system using an optical see-through head-mounted display, which he named the Sword of Damocles [1]. In 1997, Ronald Azuma authored the first report on augmented reality, proposing a definition of AR that has since been widely accepted. This definition encompasses three characteristics: the combination of real and virtual; real-time interaction; and 3D registration [2]. Since then, due to the unique capability of AR technology to effectively present virtual information alongside the real environment, the technology has experienced rapid development. To date, AR can be found in almost all human activity domains, for example, safety [3], medicine [4], construction [5], education [6], and space exploration [7], and so on.

Display is both the starting point and the final destination of augmented reality technology. The quality of the integration between the virtual and real worlds is ultimately judged by the visual outcome. Research into AR display technology has never ceased.

Hua from the University of Arizona and Javidi from the University of Connecticut combined freeform surface technology with micro integral imaging display methods to achieve the magnification of virtual 3D images and the folding of optical paths using multiple freeform surfaces [8]. In 2016, Wang Qionghua's team from Sichuan University also built an augmented reality 3D display system using a semi-transparent and semi-reflective technology [9]. GANG LI and others from Seoul National University in South Korea proposed the use of holographic optical elements to achieve AR display [10]. In 2018, Javidi's team [11] proposed an augmented reality 3D display with a large depth of field based on a tunable lens. In paper [12], the study which focuses on the application of color perspective technology in virtual reality (VR) and augmented reality (AR) camera systems, aiming to explore and optimize the implementation methods of color perspective in these two fields is done.

Display-related issues are of paramount importance in AR technology. This paper compares two display schemes for augmented reality, discusses the respective strengths and weaknesses of each, and looks forward to possible solutions. It is hoped that this paper can contribute to the advancement of augmented reality technology.

2. TWO DIFFERENT DISPLAY SCHEMES

Based on the different methods of perspective and fusion, augmented reality imaging technology can be divided into optical see-through (OST) and video see-through (VST) methods. These two display schemes each have their own characteristics and are applied in various fields [13-15].

2.1 OST (Optical See-Through)

The optical see-through method is composed of microdisplays (such as micro organic light-emitting diodes, silicon-based liquid crystals, digital light processing, etc.) and optical combiners (which can be approximated as special optical lenses that can change the path of light propagation, such as freeform surfaces, array waveguides, diffractive waveguides, etc.), and its form is similar to ordinary glasses. Users can see the real world through the optical combiner with a certain light transmittance, while the microdisplay module projects virtual content into the user's field of view, allowing users to see both the real world and virtual content simultaneously, as shown in Figure 1.

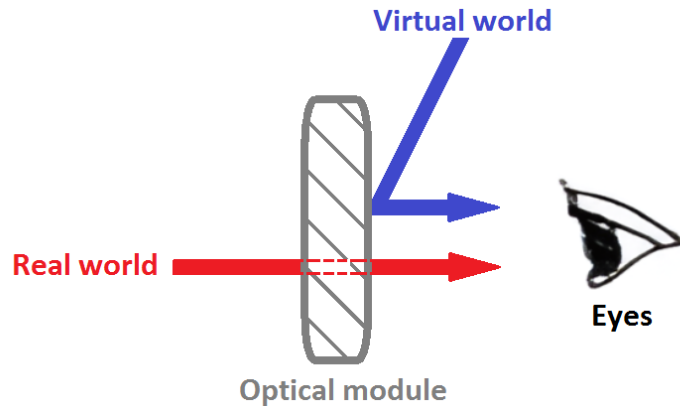


Figure 1. Schematic diagram of optical see-through

The experience can be understood as wearing optical lenses to view the external world, where light from the external world directly passes through the glasses lenses into the eyes. The real world is seen through a semi-transparent optical combiner placed in front of the user's eyes. The optical combiner is also used to reflect computer-generated images into the user's eyes, thereby combining the real world with the virtual world. A representative manufacturer and product is Microsoft's HoloLens. Figure 2 shows the HoloLens glasses. The development of Microsoft HoloLens glasses originated from Windows Holographic and has been iterated to the second generation "HoloLens 2" on November 7, 2019.



Figure 2. Microsoft HoloLens 2

2.2 VST (Video See-Through)

The video see-through method utilizes optical camera technology to capture the actual scene as a video stream, and overlays virtual content onto the scene video stream to achieve the fusion of reality and virtuality. The real world is captured in real-time by a camera, which is then combined with computer imagery technology. The camera-captured "real world" is overlaid with computer-generated "virtual images" and presented together on an opaque display, achieving a transparency-like effect, as shown in Figure 3. Representative manufacturers and products include Apple's Vision Pro, see Figure 4. Apple's Vision Pro is the first headset released by Apple inc. on June 6, 2023. It has multiple cameras and users can control it with gestures, eyes, or voice commands.

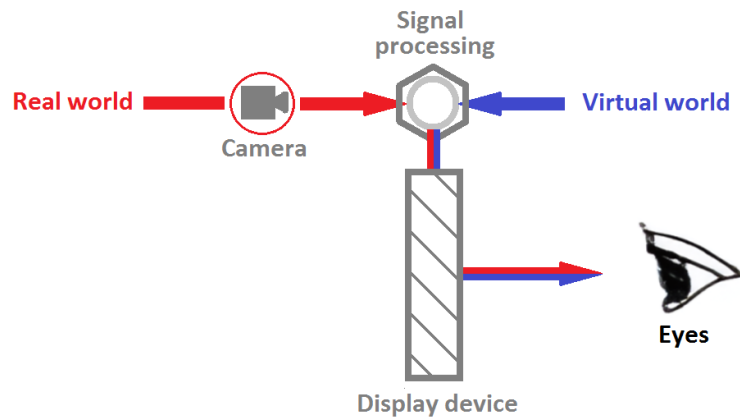


Figure 3. Schematic diagram of video see-through



Figure 4. Apple's Vision Pro

3. PERFORMANCE COMPARISON OF TWO DISPLAY SCHEMES

OST and VST, based on their respective imaging principles, exhibit distinct characteristics in their display effects.

3.1 Real-world resolution

In the OST scheme, the image of the real world directly enters the eye, hence the display resolution of the real world is determined by the human retina, with a single eye resolution of approximately 24k+. In the VST scheme, the image of the real world is a captured video, thus the display resolution of the real world is limited by the resolution of the display screen, with a single eye resolution of approximately 2k-4k.

In this field, the OST scheme is about 10 times that of VST, the OST scheme has obvious advantages.

3.2 Image quality

In the OST scheme, since the displayed images do not involve digital images of the real world, any distortion in the displayed images must be corrected optically, which increases the design complexity and cost of optical components, as well as the weight of the headset. In the VST scheme, image distortion can be eliminated through digital image processing methods, allowing for a larger field of view at a relatively low cost while achieving better image quality; however, the capture and presentation of the real environment image are limited by the camera's performance parameters, which may lead to issues such as overexposure, excessive noise, loss of focus, and jitter.

In this field, OST and VST have their own strengths.

3.3 Field of View (FOV)

Due to the limitations of optical components and display devices, along with the demand for lightweight design, the mainstream OST solution's FOV is around 30-70°. The VST solution uses larger display screens and can place the display and optical devices directly in front of the eyes, thus achieving a larger FOV, which is currently between 90-120° in the mainstream.

In this field, the advantages of VST are quite obvious.

3.4 Occlusion effect

In the OST scheme, virtual objects do not completely occlude real objects, making the virtual objects appear translucent, like ghosts, superimposed on real objects, or the boundaries between virtual and real objects are difficult to handle with pixel-level precision. In the VST scheme, since information is provided in the form of digital images, the VST combiner can simulate transparency by using real, virtual, or a mix of both on a pixel-by-pixel basis, thus producing more convincing occlusion effects than OST.

One of the main shortcomings of OST is the inability to solve occlusion problem effectively.

3.5 Brightness and brightness matching control

The OST scheme is greatly affected by the brightness of the real environment. Its display brightness is determined by the scheme's transmittance and the brightness of external light, which can reach 6600 nits or higher, more than 10 times the requirement of current VST. This directly leads to the OST scheme being limited by the display module; if the real environment is too bright, the real world will wash out the virtual image; if the real environment is too dark, the virtual image will dilute the real world.

In the VST scheme, since the real environment is captured by a camera and displayed on the screen, the brightness of the external environment seen will be limited by the screen brightness, which is currently between 100-600 nits. However, because both the real and virtual images in VST are generated by the monitor, both can undergo some dynamic range adjustment, thus making it more convenient to perform brightness matching control.

In this field, VST is clearly less restricted by external factors and has more flexible choices.

3.6 Latency

OST provides a real-time view of the real world, but there is always some latency between the virtual view and the real world, which can lead to alignment issues and affect the user experience. For VST, the latency of the real-world view captured by the camera can be delayed to match the latency of the virtual view. In other words, VST provides a method to reduce or avoid the temporal mismatch between the captured real-world view and the virtual image, but there is still a fairly consistent latency between the two and the physical real world.

OST and VST both have latency, but the mechanisms behind their occurrence are completely different.

3.7 Focal plane

In the OST scheme, the real world seen through the optical module has an infinite number of focal planes (allowing one to focus on distant or close objects at will), which can prevent the occurrence of dizziness. In the VST scheme, due to the hardware display of the optical module, there is currently only 1 focal plane.

In this field, the advantages of OST are quite obvious.

3.8 Registration information

In the OST scheme, the system's only source of information about the user's head position is the head tracker. The VST scheme provides another source of information: digital images of the real scene, which means that VST can detect features in the images and use these features to increase the accuracy of registration.

Although VST scheme can provide an additional source of information, it also requires higher computing power.

4. CONCLUSION

OST and VST, as the current two primary AR imaging schemes in augmented reality, each have their own strengths and weaknesses.

The optical see-through method's advantage lies in the user's ability to directly see the real scene without distortion of the real world. Its shortcomings include poor accuracy in the registration of virtual and real objects, poor fusion immersion, inability to handle occlusions between virtual and real objects, limited field of view, and inability to scale. The OST scheme is advantageous in conditions where the positional relationship between virtual information and the real environment is relatively simple, the real environment itself is relatively uniform, but there is a need for rapid, large-range field of view changes and interaction with the real environment.

The video see-through method's advantages include high precision in the registration of virtual and real objects, good fusion immersion, effective handling of occlusions between virtual and real objects, a large field of view, scalability, and adjustable brightness. Its shortcomings lie in the need to process the real and virtual images synchronously, which may result in some latency depending on the imaging frame rate, processing complexity, and speed. This method is more suitable for occasions where the virtual information and the real environment are relatively rich and the positional relationships are complex, but rapid movement is not required.

OST excels over VST in terms of environmental display fidelity and image processing speed, but its display effects are greatly affected by the natural environment, especially brightness, and it struggles with handling occlusions between virtual and real objects. Resolving the matching of virtual content with the natural environment and the occlusion relationships resulting from spatial positional relationships are key to achieving breakthroughs for OST display schemes in future practical applications.

VST outperforms OST in occlusion handling, field of view, latency, registration information, and brightness matching control. However, these advantages of VST all stem from its modifiability of two signals, which places extremely high demands on the system's algorithms and computing power. Higher-performance chips, more effective algorithms, and ultra-high-resolution displays are key to the advancement of VST schemes.

REFERENCES

- [1] Carmigniani J, Furht B, Anisetti M, et al. Augmented reality technologies, systems and applications. *Multimedia Tools and Applications*, 2011, 51(1): 341-377.
- [2] Shi X G, Xue Z H, Li H H, Wang B J, Li S L. Review of augmented reality display technology. *Chinese Optics*, 2021, 14(5): 1146-1161.
- [3] Gregory Kipper. *Augmented Reality*. Elsevier, Inc. 2013
- [4] Yevgeniya D, Madina I, Dana T. Towards Metahospital: augmented and virtual reality in medicine. *Procedia Computer Science*, 2024, 231: 373-378.
- [5] Maha A, Wael A A. An integrating contextual approach using architectural procedural modeling and augmented reality in residential buildings: the case of Amman city. *Heliyon*, 2022, 8: e10040.
- [6] Bernardo Marques , Beatriz Sousa Santos, Paulo Dias. Ten years of immersive education: Overview of a Virtual and Augmented Reality course at postgraduate level. *Computers & Graphics*, 2024, 124: 104088.
- [7] Flavie A. A. S. D. T. Rometsch, Andrea E. M. Casini, Anne Drepper, etc. Design and evaluation of an Augmented Reality tool for future human space exploration aided by an Internet of Things architecture. *Advances in Space Research*, 2022, 70: 2145-2166
- [8] Hua H, Javidi B. A 3D integral imaging optical see-through head-mounted display. *Optics Express*, 2014, 22(11): 13484-13491.

- [9] Deng H, Wang Q H, Xiong Z L, et al. Magnified augmented reality 3D display based on integral imaging. *Optik-International Journal for Light and Electron Optics*, 2016, 127(10): 4250-4253.
- [10] Li G, Lee D, Jeong Y, et al. Holographic display for see-through augmented reality using mirror-lens holographic optical element. *Optics Letter*, 2016, 41(11): 2486-2489.
- [11] Shen X, Javidi B. Large depth of focus dynamic micro integral imaging for optical see-through augmented reality display using a focus-tunable lens. *Applied Optics*, 2018, 57(7): B184-B189.
- [12] Zhang H Y. Research on Color Perspective of VR and AR Camera System. *IT REPORT*, 2024, 40(6): 167-169.
- [13] Lucia C, Chiara P, Stefano R. A method for user-customized compensation of metamorphopsia through video see-through enabled head mounted display. *Pattern Recognition Letters*, 2021, 151: 252-258.
- [14] Daniel P, Feng Z N, Maddy K. Optical see-through augmented reality fire safety training for building occupants. *Automation in Construction*, 2024, 162: 105371.
- [15] Fahmi B. Use Case of the Application of the Optical-See-Through Augmented Reality Technology in the Engineering Context. *Procedia CIRP*, 2022, 106: 3-8.